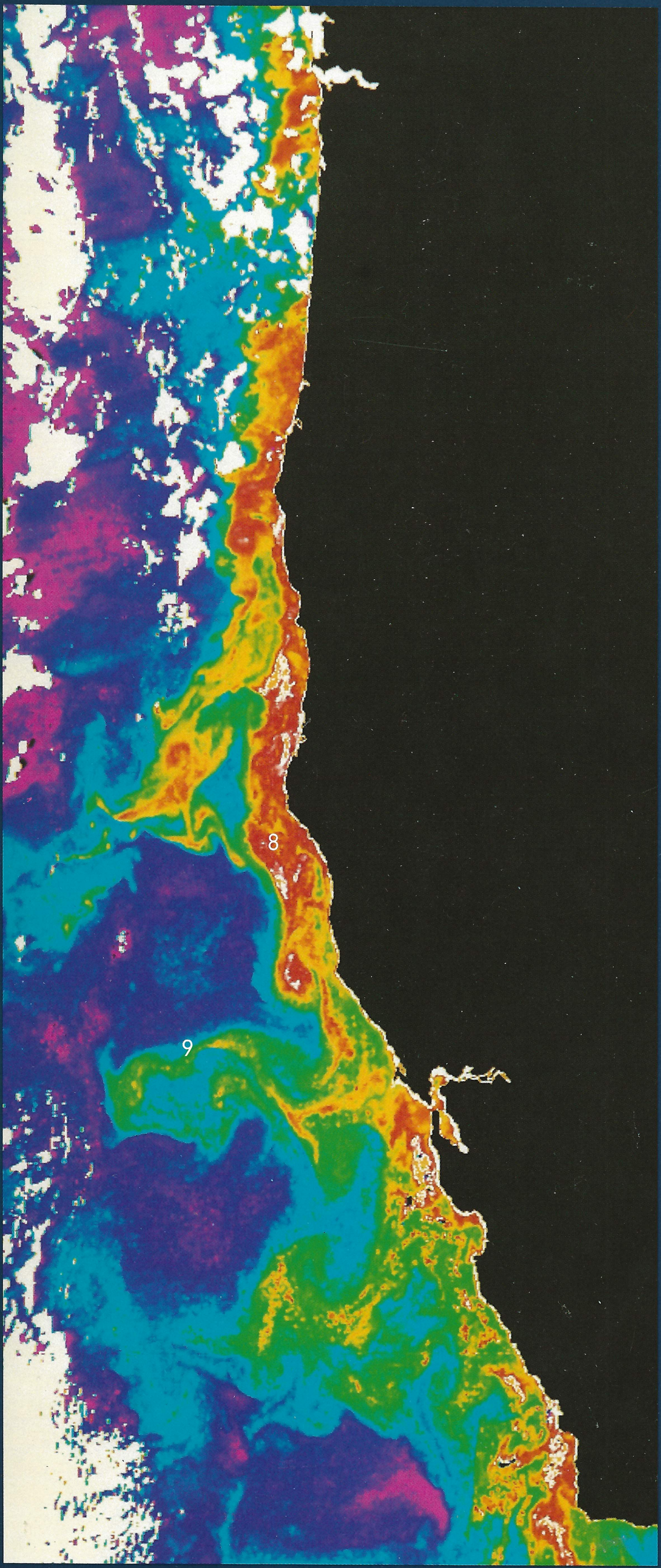
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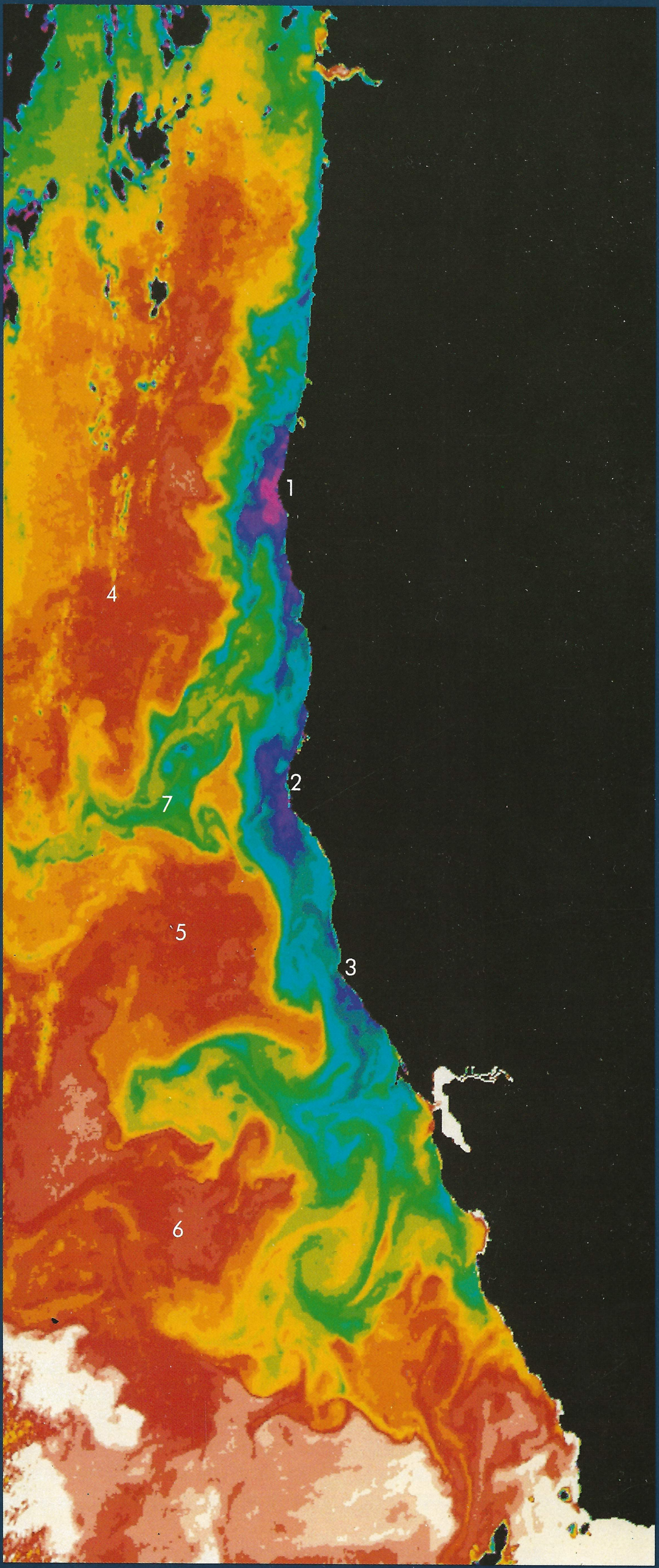
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West Coast Upwelling

The oceans are highly variable on all time and space scales. Research vessels allow us to obtain important data by direct observations, but their relatively slow speed limits the extent of shipboard sampling, making the assessment of larger-scale oceanic processes extremely difficult. For example, during the many weeks required for a ship to sample the area off the west coast of the US seen here, oceanic conditions change markedly and make it impossible to observe the overall pattern in question. In contrast, satellite sensors can provide detailed observations of the same area in only ten minutes, giving an essentially simultaneous snapshot of that pattern.

These satellite images, taken eight hours apart on July 8, 1981, represent sea-surface temperature (right) and phytoplankton chlorophyll pigment concentration (left). The sea-surface temperature image was derived from infrared temperature readings from the Advanced Very High Resolution Radiometer (AVHRR) on board the NOAA-6 meteorological satellite. The chlorophyll image was derived from ocean color measurements from the Coastal Zone Color Scanner (CZCS) on NASA's Nimbus-7 satellite. (See: Western North Atlantic Color.) Data were received at the Satellite Oceanography Facility at the Scripps Institution of Oceanography and processed at the Jet Propulsion Laboratory by Mark Abbott and Phil Zion, using computer processing routines developed at the University of Miami.

In the summer, prevailing winds from the northwest drive coastal surface waters offshore, inducing an upwelling of cooler subsurface water. In the temperature image (right) the cool, freshly upwelled water along the coast is shown in violet-purples (8 to 9° C or 47 to 49° F), especially noticeable at Cape Blanco (1), Cape Mendocino (2), and Point Arena (3). Intermediate temperature water is indicated by blues and greens, and the warmer California Current water offshore is depicted in yellows and reds (14 to 15° C or 57 to 59° F).

Several large meanders of the California Current are visible (4,5,6), as well as long filaments of cool upwelled water (7) which extend 100 to 300 km (60 to 180 mi) offshore. Based on half a century of ship observations, the California Current was thought to be broad and slow, with a uniform flow to the south. In contrast, analysis of satellite data shows a meandering current which interacts in a complex way with the coastal waters to produce the swirls, jets, and eddies apparent in these images.

As phytoplankton and their associated chlorophyll pigments become more abundant, ocean color shifts from blue to green. This color shift is detected by the CZCS sensor and used to calculate chlorophyll pigment concentration. The chlorophyll image (left) shows high levels of phytoplankton along the coast in reds (chlorophyll concentration greater than 10 mg/m^3), intermediate levels are shown in yellows and greens, and the lowest levels offshore (less than 0.1 mg/m^3) are purples.

The cool, subsurface water upwelled along the coast is rich in nutrients. As it spreads out into the sunlit surface layer, it provides an ideal environment for phytoplankton growth. This results in the high levels of phytoplankton (8) indicated by reds in the image. Satellite data have shown for the first time that much of the phytoplankton growth in the coastal zone is entrained into jets or filaments and carried far offshore (9). This new discovery has an important implication: Since phytoplankton levels are about 100 times higher along the coast than in the nutrient-poor offshore waters, this transport of phytoplankton is the most likely food source for the high zooplankton populations observed several hundred kilometers from the coast.

Coastal upwelling systems such as this one are among the most productive regions of the world ocean. Some phytoplankton settle to the bottom, but most are eaten by zooplankton and fish, with some of the fish subsequently being harvested by man. Many of the great fisheries of the world are in upwelling areas including those along the west coast of North America, the west coast of Peru and Ecuador, and the northwest and west coasts of Africa.

Analysis of long-term satellite data sets will provide insight into the seasonal and interannual variability of these upwelling systems, enabling a better understanding of associated changes in fish stocks and potentially more intelligent management of the fisheries.

The color and temperature patterns in these images, both very similar, reflect the current patterns present in this area. Such images provide a means for flow visualization, useful not only for scientific studies but also for operations associated with the offshore petroleum and deep-sea mining industries. These operational activities are critically dependent on knowing when and where strong currents, such as the offshore jets seen in these images, might occur.

The AVHRR will continue to be carried as an operational sensor on the NOAA meteorological satellites through the 1990's. The CZCS is a demonstration instrument launched in 1978 with a design life of one year. Remarkably it has survived over seven years, but it is now beginning to fail. An improved version, the Ocean Color Imager (OCI), has been proposed as a follow-on instrument. Beyond the OCI, NASA is developing a new type of instrument, the Moderate Resolution Imaging Spectroradiometer (MODIS). The CZCS and OCI measure the light reflected from the ocean in a few selected color bands; in contrast, MODIS measures the whole visible and much of the infrared spectrum in many narrow bands. MODIS offers a new measurement capability; for example, it may be possible to identify dominant phytoplankton groups and to distinguish between different materials suspended in the water. Also useful for a variety of land and atmospheric measurements, MODIS is part of an Earth Observing System (Eos) planned for the mid-1990's and directed toward looking at the entire Earth as an integrated system.

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